

Acute Motor Axonal Neuropathy: An Antibody-mediated Attack on Axolemma

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The acute motor axonal neuropathy (AMAN) form of the Guillain-Barré syndrome is a paralytic disorder of abrupt onset characterized pathologically by motor nerve fiber degeneration of variable severity and by sparing of sensory fibers. There is little demyelination or lymphocytic inflammation. Most cases have antecedent infection with *Campylobacter jejuni* and many have antibodies directed toward GM1 ganglioside-like epitopes, but the mechanism of nerve-fiber injury has not been defined. In 7 fatal cases of AMAN, immunocytochemistry demonstrated the presence of IgG and the complement activation product C3d bound to the axolemma of motor fibers. The most frequently involved site was the nodal axolemma, but in more severe cases IgG and C3d were found within the periaxonal space of the myelinated internodes, bound to the outer surface of the motor axon. These results suggest that AMAN is a novel disorder caused by an antibody- and complement-mediated attack on the axolemma of motor fibers.

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In the postpolio era, the Guillain-Barré syndromes (GBSs), considered together, are the most frequent cause of acute flaccid paralysis worldwide. GBSs usually follow an infectious illness, and in most cases the pathogenesis appears to reflect an inflammatory autoimmune attack on the myelin-forming cells of the peripheral nervous system, resulting in demyelination [1, 2] (for review, see reference 3). In some cases the target of immune attack has been proposed to be the axon, resulting in degeneration of motor and sensory axons [4–7]. Recently, a novel form of GBS, termed acute motor axonal neuropathy (AMAN), has been recognized in large summer epidemics in northern China [6, 8–10] and occurs, as well, in sporadic cases in Japan, Korea, India, Mexico, Europe, and North America [11–18]. AMAN is a noninflammatory disorder that, in severe cases, produces selective degeneration of the axons in the motor roots and nerves without demyelination [6, 8–10, 12, 15, 16].

AMAN often follows infection with the enteric bacterial pathogen *Campylobacter jejuni* [8, 10, 17, 18]. Some evidence suggests that specific *C. jejuni* strains may be involved [19, 20], that at least some of these *C. jejuni* isolates contain in their lipopolysaccharides the moiety Gal(β1–3)GalNAc, an epitope present in GM1 ganglioside, and that AMAN patients have an increased likelihood of having antibodies, particularly

of the IgG class, that bind to GM1 [15, 18, 21, 22]. The Gal(β1–3)GalNAc epitope is known to be present at the nodes of Ranvier of myelinated nerve fibers [23–26]. Thus, an attractive possibility for pathogenesis has been that *C. jejuni* infection in some fashion stimulates an immunologic response to this Gal(β1–3)GalNAc epitope, resulting in an immune attack on nerve fibers containing that epitope [18, 27–29]. However, as recently emphasized [3, 17, 22, 30], the pathogenetic relevance of these antibodies is uncertain. To date no direct evidence has demonstrated antibody binding to nerve fibers in tissue from patients with AMAN.

In this study, we have asked whether immunoglobulin binding and complement activation occur on nerve fibers from AMAN patients and, if so, where the deposits are localized; when in the course of disease the complement activation occurs; and what the consequent pathological changes might be. These results implicate in the pathogenesis of AMAN antibodies that are directed against epitopes that are located on myelinated motor axons, and specifically concentrated in the axolemma.

Materials and Methods

Material

Peripheral nerve samples were obtained at autopsy from 7 early cases of AMAN. All died at the Second Teaching Hos-

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Table 1. Patient Summaries

	Case ^a	Age (yr)	Sex	Symptom Onset to Death (days)	Electrodiagnostic Pattern
Extensive Wallerian-like degeneration of root or fibers	11 ^b	1.5	F	5	Reduced motor amplitudes, normal sensory studies
	2 ^b	14	M	7	Not done
	8 ^b	56	M	12	Reduced motor amplitudes
Milder pathology	10 ^b	22	M	4	Not done
	3 ^b	28	F	8	Not done
	1 ^b	9	F	9	Not done
	13	50	M	9	Reduced motor amplitudes

^aCase numbers correspond to those in reference 6.^bThe pathology has previously been described in reference 6.

Table 2. Antibodies for Immunocytochemistry

Antibody	Directed Against	Supplier	Dilution (Paraffin Sections)	Dilution (Teased Fibers and Plastic Sections)	Secondary Antibody	Blocking Serum
HAM-56	Activated macrophage	Enzo	1:100	1:25	Mouse IgM	Goat
HLA-DR	MHC class II	Dako	1:100	1:40	Mouse IgG	Horse
C3d	A complement activation product	Dako	1:2,000	1:2,000	Rabbit IgG	Goat
C5b-9	The terminal complement complex	Quidel	1:500	1:200	Mouse IgG	Horse
Biotinylated anti-immunoglobulin	IgM (μ)	Sigma	1:50	1:4	—	Goat
	IgG (γ)	Sigma	1:50	1:4	—	

MHC = major histocompatibility locus.

pital of Shijiazhuang, China (Table 1). The interval from onset of neuropathy until death was 4 to 12 days and the postmortem interval ranged from 2.5 to 10 hours. The autopsies were carried out by a protocol developed as part of an ongoing collaborative effort involving investigators at the Second Teaching Hospital, the University of Pennsylvania, and the Johns Hopkins University [6, 9]. At the time of autopsy, careful laminectomies were performed by a dorsal approach to reduce traction on the spinal roots during their removal. The samples included several sites in the thoracic and lumbar spinal cord, the dorsal and ventral spinal roots, the dorsal root ganglia, and sciatic and sural nerves. Material in a few of the cases also included median, ulnar, tibial, and peroneal nerves. At each peripheral nervous system (PNS) site, samples were collected in 4% paraformaldehyde and in 5% glutaraldehyde, as previously described [2, 6]. The former fixation was used for teased fiber and paraffin section immunocytochemistry, and the latter for immunocytochemistry on plastic-embedded sections and for electron microscopy.

Immunocytochemical Methods

PARAFFIN SECTIONS AND TEASED FIBERS. Teased fibers and paraffin-embedded tissue were prepared from specimens fixed in 4% paraformaldehyde in 0.1 M phosphate-buffered saline (PBS). For paraffin, the spinal roots and nerves were embedded in longitudinal and transverse orientations. The samples were stained for immunoglobulins, the complement activation products C3d and C5b-9, and for macrophages, using the antibodies listed in Table 2. The paraffin sections

were treated with hydrogen peroxide to remove the osmium and to block endogenous tissue peroxidase, with nonfat milk and serum to block nonspecific proteins, and with Triton X to permeabilize membranes. They were incubated overnight with specific primary antibodies, rinsed in PBS, and incubated sequentially with biotinylated secondary antibody, avidin-biotin horseradish peroxidase complex (Vector), and filtered 2,3-diaminobenzidine (DAB) (Polyscience) with hydrogen peroxide. Immunostaining of the teased fiber preparations of the spinal roots and nerves was as previously described [31, 32]. Examination of the immunostained teased fibers under Nomarski optics permitted visualization of immunoreactive cells as well as delineation of the axon, its ensheathing myelin, and the nodes of Ranvier.

For localization of complement activation products, we placed special reliance on the appearance of C3d. The emphasis on this marker results from its particular advantages for high-resolution immunolocalization. C3d is a degradation product of C3b, the biologically active fragment of C3 generated as a result of complement activation by either the classic or alternate pathway. It binds covalently to membrane at the site of activation [33, 34].

EPON-EMBEDDED SAMPLES. The tissue samples for plastic embedding were fixed in 5% glutaraldehyde in 0.1 M sodium phosphate buffer (PBS) at pH 7.4, postfixed in 2% osmium tetroxide, dehydrated in graded alcohols, and embedded in Epon in both longitudinal and transverse orientation. One-micrometer-thick plastic sections were either stained with toluidine blue or immunostained. For immuno-

cytochemistry we used a modification of the method of Trapp and colleagues [35]. The plastic sections were etched in a saturated sodium methoxide solution in propylene oxide for 5 minutes, rinsed in absolute alcohol, and treated for 10 minutes in 1% sodium borohydride [2]. The etched sections were immunostained by the avidin-biotin peroxidase method described above. The antibody dilutions for the plastic sections are specified in Table 2.

Thick/thin pairs of sections were prepared by obtaining a 1-micrometer-thick plastic section for immunocytochemistry and taking the next section (ie, with no intervening thick or thin sections) as a thin section for electron microscopy [35]. The thin sections were mounted on copper mesh grids and stained with lead and uranyl acetate for examination by electron microscopy. By this method, selected immunostained fibers were examined ultrastructurally. The results in the motor roots and mixed spinal nerves of these cases were compared with those in the unaffected dorsal roots, as well as with control nerves from normal sudden-death controls. In separate reports we also describe the findings in the acute inflammatory demyelinating form of GBS [2] and in a series of disease controls, including sural nerve biopsies from patients with ongoing Wallerian-like degeneration due to sensory ganglion cell destruction or to vasculitis (Sheikh K, Hafer-Macko C, Hsieh S-T, Ho TW, and Griffin JW, unpublished data).

Results

All 7 AMAN patients died between 4 and 12 days after onset of the first neuropathic symptoms, with an 8-day median interval. The clinical characteristics and the pathologic characterization of these cases has previously been reported [6]. The pathologic studies included teased fibers, paraffin-embedded sections, plastic sections, and ultrastructural analysis [6, 9]. Three of these cases (nos. 2, 8, and 11) had extensive Wallerian-like degeneration extending from the ventral root exit zone and involving motor fibers nearly exclusively. In 4 cases (nos. 1, 3, 10, and 13), the pathology at the light microscopic level was mild and was restricted to a small proportion of degenerating fibers in the ventral roots and peripheral nerves [6, 8]. At the ultrastructural level, many other motor fibers in these cases had lengthening of the nodes of Ranvier [9] and a distinctive picture of macrophages within the periaxonal spaces, surrounding a condensed-appearing axon [6, 9]. As previously reported, demyelination was infrequent [6]. In some instances, it appeared to be a consequence of nodal axonal enlargement and consequent displacement of the paranodal myelin [9]. However, occasional examples of macrophage-mediated demyelination, comparable with those found in acute inflammatory demyelinating polyneuropathy (AIDP), could be found in Cases 2 and 11 [6].

Nodal Changes

IMMUNOGLOBULIN/COMPLEMENT STAINING. The outstanding finding was the intense immunoreactivity for

the complement activation product C3d at the nodes of Ranvier of many motor fibers. The discrete localization of C3d at the nodal axolemma was most clearly evident on teased fibers (Fig 1A–D). Immunostaining was present on the lengthened nodes [9] more frequently than on normal nodes. Neither small myelinated fibers of the ventral root (γ -efferent fibers) nor the fibers in the dorsal root had C3d immunostaining. The C3d immunoreactivity of the nodes of large fibers in the ventral root nodes was also apparent in plastic sections (Fig 2A). Electron microscopy of these same immunostained nodes (Fig 3A) confirmed that the C3d deposition was present on the nodal axolemma, rather than the Schwann cell, its microvilli, or macrophages. In general, C3d deposition was most prominent in the earliest cases (nos. 10, 11, and 12) and in fibers that had not begun Wallerian-like degeneration. Degenerating fibers did not immunostain for C3d.

In the plastic sections, some nodes were also stained for the terminal complement complex C5b-9, and some nodes also had intense immunoreactivity for IgG. However, the normal diffuse IgG staining that is due to the presence of immunoglobulin within the endoneurial fluid made it difficult to determine what proportion of the nodes had specific, rather than “background,” immunoreactivity.

MACROPHAGE STAINING. Many ventral root fibers that had not begun Wallerian-like degeneration had macrophages overlying the nodes of Ranvier. They were present in 54% of the nodes of Ranvier of large myelinated ventral root fibers in immunostained teased fibers preparations of the 4-day case (Fig 1E–H). Macrophages overlying nodes were also appreciated on plastic sections (Figs 2A, 3A, B). They often extended processes through the Schwann cell basal lamina to contact the nodal axolemma (see Figs 1E, G, H; 3A, B). They were present primarily over the lengthened nodes of large myelinated fibers but were occasionally found over normal-appearing nodes (see Fig 1B). They were not seen over the nodes of small myelinated motor fibers or over dorsal root fibers in these AMAN cases.

Internodal Changes

IMMUNOGLOBULIN/COMPLEMENT STAINING. The internodal axolemma of a population of myelinated motor fibers was intensely immunostained for IgG (see Fig 2B) and C3d (see Fig 2C). As with the nodal changes, this pattern of staining anteceded Wallerian-like degeneration. In some fibers, the terminal complement complex C5b-9 was also identified in the internodal periaxonal space. Many of these immunostained axons were condensed and displaced away from their overlying myelin sheath (see Fig. 2B). Such axons showed clearly that the staining was on the surface of the axon and not on the surrounding Schwann cell

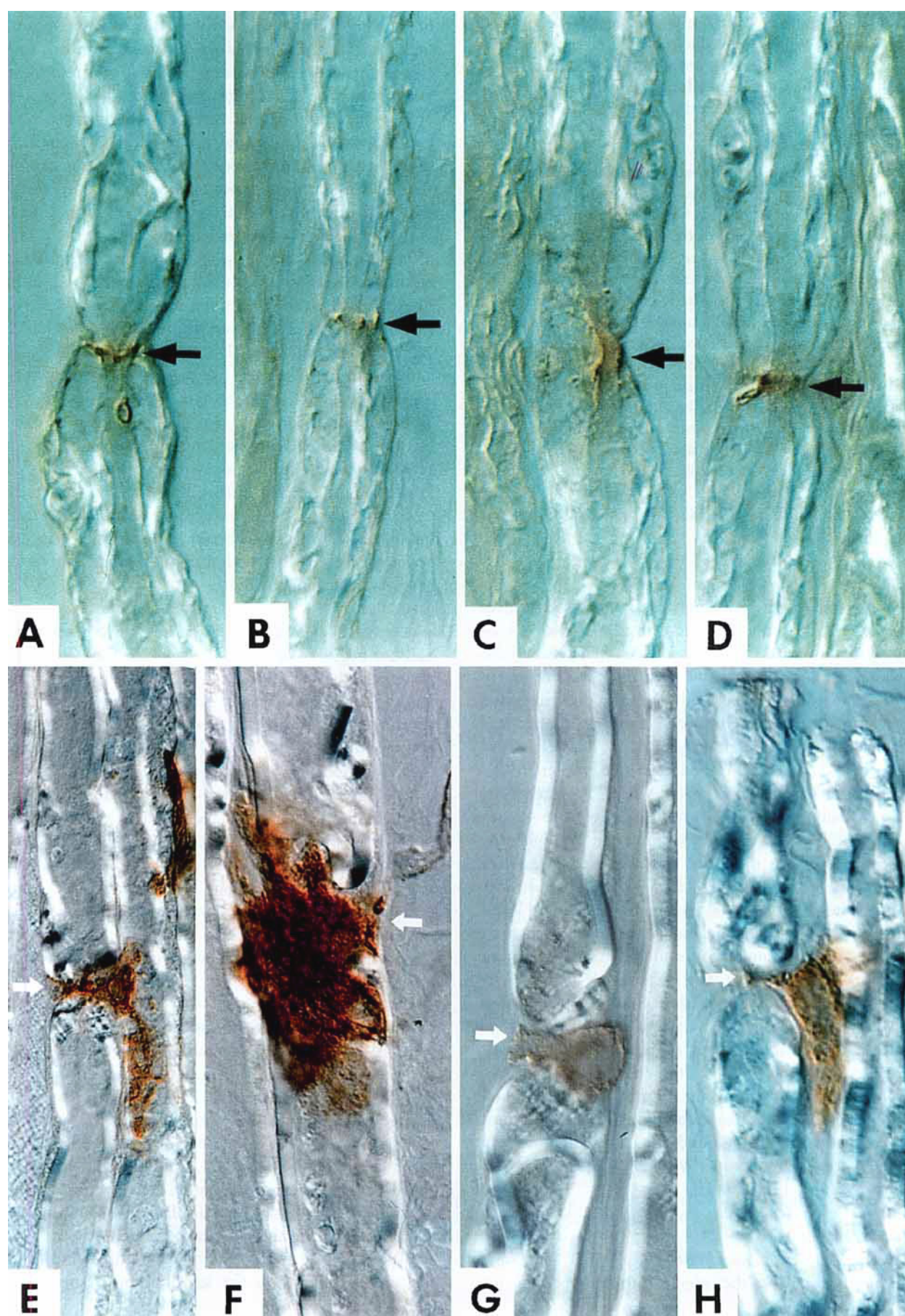


Fig 1.

or the inner aspect of the myelin sheath. Within the perimeter formed by the intensely stained axolemma, the axoplasm of these fibers often appeared to be lightly immunostained by antibodies against IgG (see Fig 2B). IgG was also present throughout the endoneurial space, but the intensity of staining was less than the axonal staining. Antibodies to IgM faintly stained the endoneurium but not the internodal axolemma.

The myelin sheaths around those axons that immunostained with IgG and C3d were normal at both the light and electron microscopic levels (see Fig 3B). C3d was never identified on the myelin sheath or on the outer Schwann cell surface, but light staining with C5b-9 was seen on the outer surface of some fibers.

MACROPHAGES. In many internodes of large motor fibers, macrophages were found inside the innermost ring of the myelin sheath. The identification of these cells as macrophages was confirmed by their immunostaining with HAM-56 and for HLA-DR (major histocompatibility locus [MHC] class II) (Fig 2E, F). They invariably stained for C5b-9 as well. In the corresponding thin sections, these macrophages were confirmed to be within the internodal periaxonal space, separating the axon from the adaxonal Schwann cell and myelin sheath (Fig 3C, D). Most of these fibers had condensed-appearing axons (see Figs 2F, 3D), but they did not damage even the innermost myelin lamella (see Fig 3D). Many of the fibers that were found at the ultrastructural level to have periaxonal macrophages also had IgG and C3d deposition on the internodal axolemma, as determined by the thick–thin comparisons.

In these AMAN cases, macrophages were never associated with the outer or abaxonal surface of the Schwann cell, as reported in AIDP [2]. There was no internodal demyelination or macrophage-mediated myelin stripping. In addition to their unusual location, macrophages in axonal cases differed from those associated with AIDP [2] in several other respects. The macrophages associated with the axonal cases were MHC class II (HLA-DR) positive but were not as large or foamy. This also contrasts with the inflammatory demyelinating pattern of GBS, in which the macrophages become filled with lipid droplets and myelin debris [2].

Control Nerves

In control nerves IgG was diffusely distributed throughout endoneurial spaces but was not seen in the

periaxonal space in either the normal control or the disease control nerves (Sheikh K, Hafer-Macko C, Hsieh S-T, Ho TW, and Griffin JW, unpublished data, 1996). In a similar manner, we did not see C3d deposition on nerve fibers in any of the normal or disease control nerves. In particular, as will be reported separately, we did not see deposition on fibers undergoing Wallerian degeneration (Sheikh K, Hafer-Macko C, Hsieh S-T, Ho TW, and Griffin JW, unpublished data, 1996). C5b-9 was present in macrophages in several disease control nerves, as it was in the present cases, but we did not encounter periaxonal macrophages in the other disorders. Finally, as noted, the pattern in nerves from individuals dying with the inflammatory demyelinating form of GBS was quite different. In particular, C3d was found on the outer surface of the Schwann cell, outside of the myelin sheath [2].

Discussion

These studies localized IgG and complement activation markers to the axolemma in AMAN. The deposition was restricted to large motor fibers, and was present at early stages of the disease, before Wallerian-like degeneration supervened. Complement activation products were present on a high proportion of the nodes of large motor fibers in the earliest case. Macrophages, probably attracted by C5a and other complement products, were recruited to the affected nodes and subsequently dissected into the internodal periaxonal spaces. The myelin sheath in such fibers usually remained intact, but the axon was displaced away from the inner Schwann cell plasmalemma by the intervening macrophage and appeared condensed. In the more severe cases many fibers eventually degenerated. Taken together, these results suggest that AMAN is likely to be an IgG- and complement-mediated disorder, and that the target epitopes are likely to be constituents of the axolemma of motor fibers.

The localization of IgG and complement activation products in AMAN differed markedly from their localization in the acute inflammatory demyelinating form of GBS. Several previous studies have suggested that complement markers in AIDP are located in association with the Schwann cell or myelin sheath. Using a similar technique of immunostaining and ultrastructural correlation, we have recently reported that C3d and C5b-9 are located on the abaxonal Schwann cell

◀ Fig 1. (A–D) The complement activation product C3d was localized discretely at nodes of Ranvier (black arrows) of large myelinated motor fibers, as demonstrated on these immunostained teased ventral root fibers from the case 4 days after the onset of neurologic symptoms. The golden-brown immunoreaction product is at the nodes. (E–H) These fibers demonstrate that many motor fibers had macrophages overlying and extending processes into the nodes of Ranvier (white arrows designate nodes of Ranvier). (E and F were immunostained with the macrophage marker HAM-56; G and H for HLA-DR [major histocompatibility locus class II].)

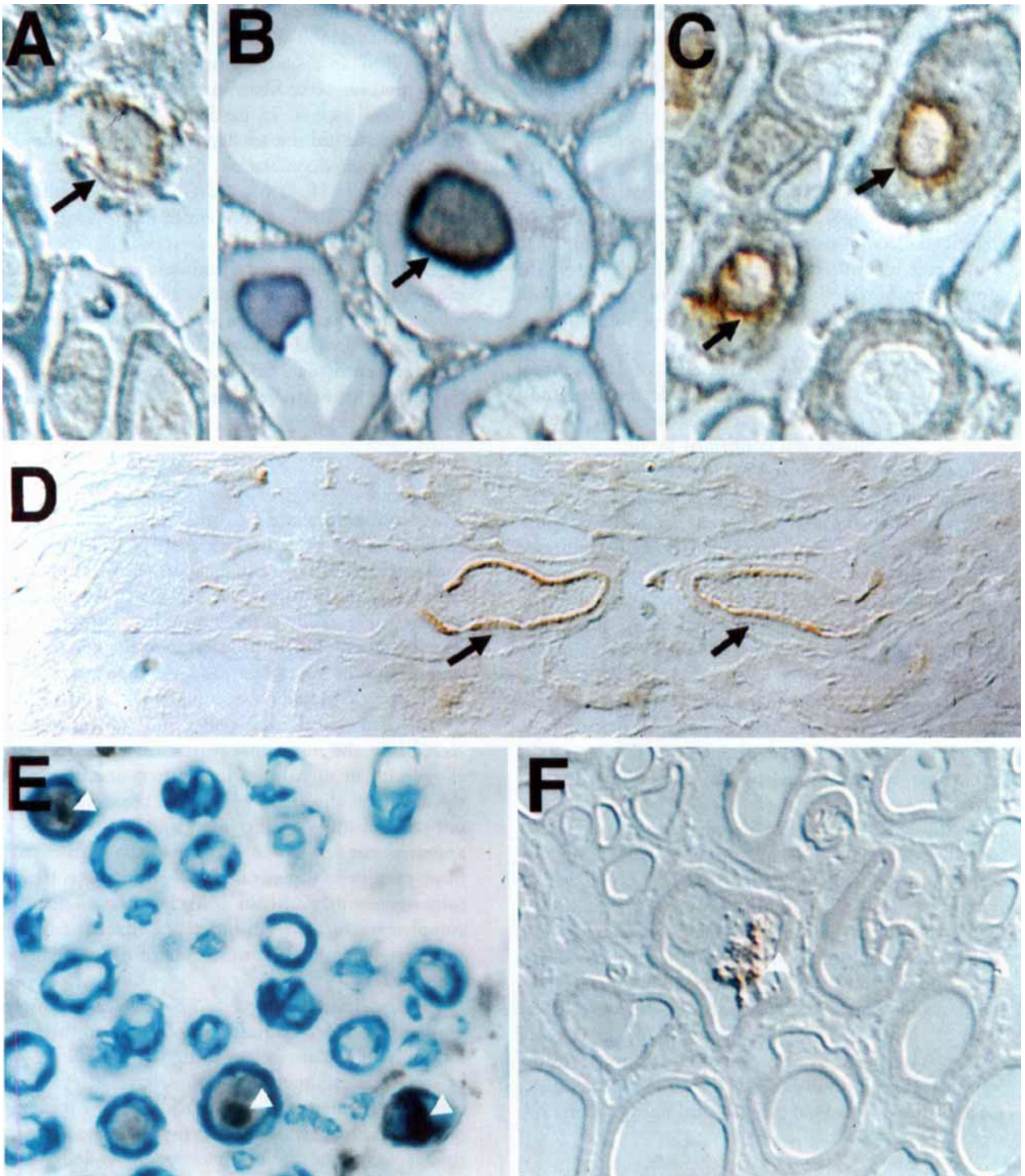


Fig 2. (A) C3d deposition was present around this node of Ranvier (black arrow) in an immunostained etched plastic section of ventral root from an acute motor axonal neuropathy case 5 days after onset of neurologic symptoms. This same node is shown at the electron microscopic level in Figure 3A. Note that the nearby cell identified by the white arrowhead is a macrophage, as confirmed in 3A. (B) IgG was deposited along the internodal axolemma (black arrow) within the periaxonal space, between the axon and myelin sheath. Less intense staining was present within the axon and throughout the endoneurium. (C and D) C3d immunoreactivity (black arrows) was also present in the periaxonal space of some myelinated internodes, as seen by transverse (C) and longitudinal (D) sections. (E and F) Macrophages (white arrowheads) eventually insinuated themselves into the periaxonal space, dissecting between the nodal axolemma and Schwann cell (E and F) in this location. Macrophages were major histocompatibility locus class II (HLA-DR) positive (E) but did not appear as large, foamy, or filled with liposomes and lipid droplets as would be expected had they been postphagocytic. Most fibers with macrophages in the periaxonal space had condensed-appearing axons (F) (stained with the macrophage marker HAM-56).

plasmalemma in early AIDP cases [2]. The patterns of complement and macrophage localization provide satisfying correlates to the axonal and demyelinating patterns of GBS, and suggest that the responsible antigens are in different sites in the myelinated fiber, ie, in AMAN, in the nodal and internodal axolemma of motor fibers, and in AIDP, in the plasmalemma of Schwann cells. So far, we have not identified transitional cases in which both patterns of localization are present. That such cases might occur is suggested by previous reports in which lymphocytic infiltration and demyelination were identified in cases with prominent axonal degeneration and periaxonal macrophages [36] (Vallat JM, personal communication, 1996).

The target antigen present in the motor fiber node is not known, but several lines of evidence suggest that the Gal(β 1–3)GalNAc epitope is a candidate. This epitope is recognized by some anti-GM1 antibodies [37], and many patients with axonal forms of GBS have anti-GM1 antibodies in their sera [16, 18, 21, 22, 37, 38]. As noted above, many AMAN cases follow infection with *Campylobacter* species. Some of the isolates of *Campylobacter* from AMAN patients have had the Gal(β 1–3)GalNAc epitope as part of a GM1 ganglioside tetrasaccharide within their lipopolysaccharides [28, 39], suggesting that the lipopolysaccharide may stimulate an immune response to GM1. The organisms isolated from other GBS patients with anti-GM1 reactivity have not had GM1-like epitopes detected in the lipopolysaccharide, and other mechanisms leading to anti-GM1 antibodies have not been excluded [3, 22, 37]. In any event, GM1-like epitopes are known to be highly concentrated in the nodes of Ranvier of myelinated fibers [23, 24, 40, 41], and Santoro and colleagues [25] have previously described nodal localization of C3d and IgM in a sural nerve biopsy from a patient with motor neuron disease and elevated anti-GM1 IgM antibody titer. There was nodal deposition of IgM following intraneural injection of this anti-GM1 patient serum into rat sciatic nerves [42]. Rabbits immunized with GM1 or its Gal(β 1–3)GalNAc epitope produced antibodies, developed conduction block, and had immunoglobulin deposited at nodes [26]. Finally, the practice of injection of mixed gangliosides as a putative treatment for a variety of disorders has been associated with the development of axonal neuropathy and anti-GM1 antibodies that bind to rat nodes and neuromuscular junctions [43].

Antibody-mediated alterations in nodal sodium channels provide a novel potential mechanism for block of impulse conduction [44]. Sera from rabbits immunized with GM1 has recently been shown to alter ion channel function at nodes, as shown by voltage-clamp studies [45]. In the presence of complement, anti-GM1 antibodies are capable of suppressing the sodium current and causing nonspecific current leakage.

In the absence of complement in this model, anti-GM1 antibodies increase the rate of rise and amplitude of the potassium current, probably by uncovering paranodal potassium channels [45]. These data suggest that simple binding of antibody, alone or with subsequent activation of complement at the nodes of motor fibers, can be sufficient to impair conduction. Assuming that such physiologic effects are reversible, they may account for the relatively rapid improvement of many AMAN cases [10]. Sera from patients with GBS and anti-GM1 antibodies [46] can produce failure of conduction in isolated mouse phrenic nerve–diaphragm preparations.

The appearance in the AMAN nerves of immunoglobulin and C3d within the periaxonal spaces of the internodes seemed to develop later than the initial nodal binding, and to involve a smaller proportion of fibers. It is likely that entrance of antibody and complement into the periaxonal space of the internode depends upon disruption of the axoglial junctions of the myelin sheath attachment sites in the paranode, allowing endoneurial fluid containing the antibody and complement to penetrate the periaxonal space and bind. Once activated in the periaxonal space, complement activation products probably act as chemotropic signals that attract macrophages to the space.

A hypothetical immunopathologic sequence of events can be constructed from the immunocytochemical results of these early AMAN cases. Enteric infection with appropriate species and strains of *Campylobacter* induces an antibody response to a glycoconjugate epitope that is found at the nodes of Ranvier of large motor fibers. For reasons noted above, GM1-reactive epitopes are an attractive candidate target, and the antibodies are likely to be isoforms of IgG that can fix complement (IgG₁ and IgG₃) [10, 21]. Antibody binding with or without complement activation may be able to alter ion channel function [45] and impair impulse conduction, potentially resulting in paralysis with little structural change and with the potential for rapid recovery [10]. Lengthening of the nodes of Ranvier, the earliest and mildest change observed [9], probably follows local activation and deposition of complement. Macrophages are recruited to the affected nodes, probably as a result of C3a, C5a, and other complement-derived chemotropic signals. The macrophages pass through the Schwann cell basal lamina that covers the node to appose the nodal axolemma [9].

These local events in the node appear to disrupt the junctional complexes between axolemma and myelin terminal loops in the paranodes that normally limit the access of macromolecules from the endoneurial space into periaxonal space. As a result, IgG and complement, among other endoneurial proteins, enter the periaxonal space of the internode. Complement activation within the periaxonal space recruits macrophages

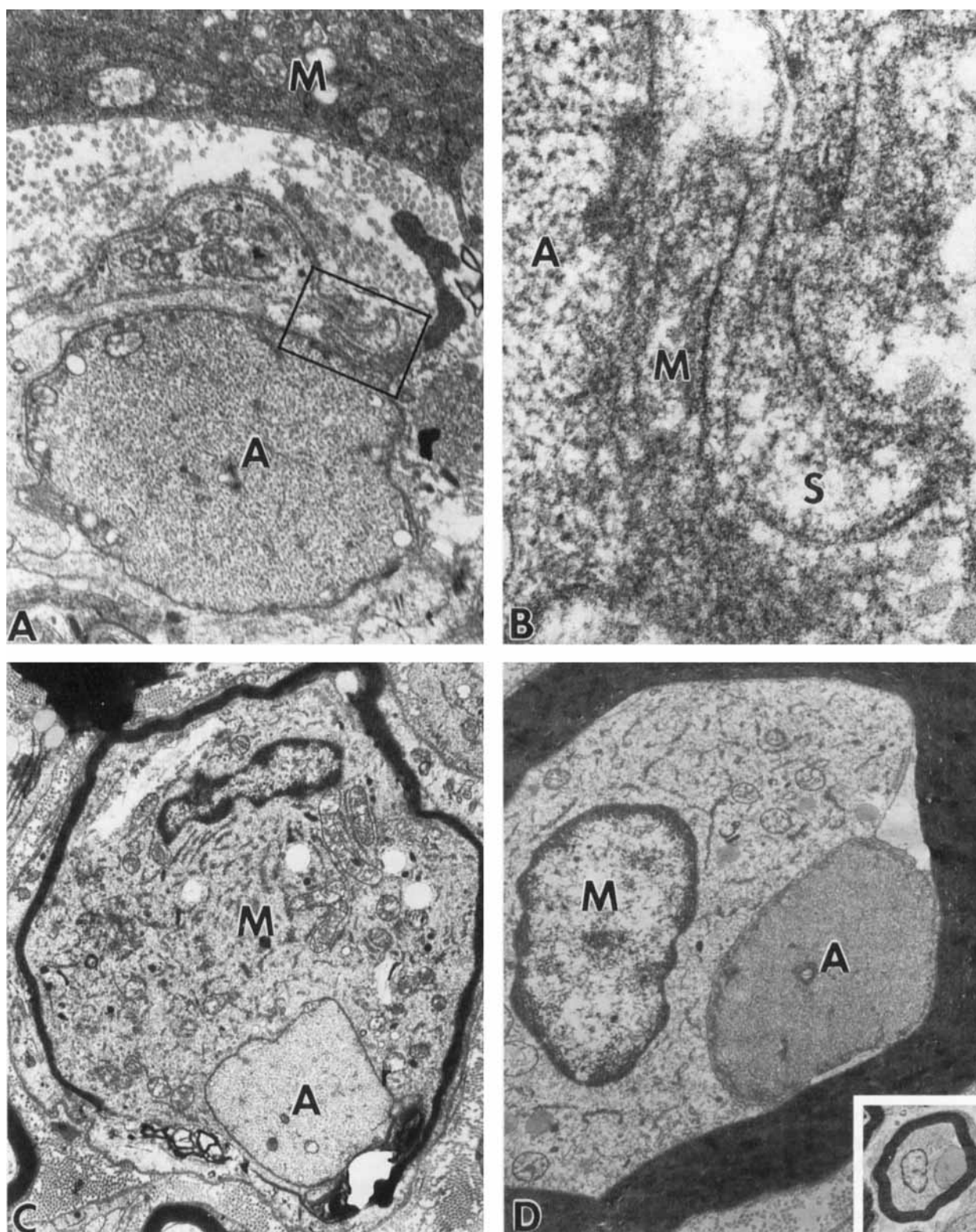


Fig 3.

into this usually inviolate site [47, 48]. Taken together, these data suggest that in AMAN the potential mechanisms that could lead to interruption of conduction in motor fibers include antibody binding at nodes, local complement activation, opening of the periaxonal space (which presumably eliminates the insulating function of myelin without damage to the myelin sheath itself), and in more severe cases, Wallerian-like degeneration of the motor axon. We have speculated that the acute motor-sensory axonal neuropathy (AMSAN) may represent the most severe degree of attack on the same epitope that is targeted in AMAN [7]. If this is the case, the epitope must be more abundant or be differently displayed in motor fibers to explain the relative vulnerability. On the other hand, motor fibers may have a unique epitope that is targeted in AMAN, and a different epitope—shared by motor and sensory fibers—may be affected in AMSAN. In either event, the present data raise the possibility that antibody-mediated immune attacks on axolemma and neuronal plasmalemmal glycoconjugates might be important in other noninflammatory axonal and neuronal degenerations.

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References

- Asbury AK, Arnason BG, Adams RD. The inflammatory lesion in idiopathic polyneuritis. *Medicine* 1969;48:173–215
- Hafer-Macko C, Sheikh KA, Li CY, et al. Immune attack on the Schwann cell surface in acute inflammatory demyelinating polyneuropathy. *Ann Neurol* 1996;39:453–463

- Hughes RA, Rees JH. Guillain-Barré syndrome. *Curr Opin Neurol* 1994;7:386–392 (Review)
- Feasby TE, Gilbert JJ, Brown WF, et al. An acute axonal form of Guillain-Barré polyneuropathy. *Brain* 1986;109:1115–1126
- Feasby TE, Hahn AF, Brown WF, et al. Severe axonal degeneration in acute Guillain-Barré syndrome: evidence of two different mechanisms? *J Neurol Sci* 1993;116:185–192
- Griffin JW, Li CY, Ho TW, et al. Guillain-Barré syndrome in northern China: the spectrum of neuropathologic changes in clinically defined cases. *Brain* 1995;118:577–595
- Griffin JW, Li CY, Ho TW, et al. Pathology of the motor-sensory axonal Guillain-Barré syndrome. *Ann Neurol* 1996;39:17–28
- McKhann GM, Cornblath DR, Griffin JW, et al. Acute motor axonal neuropathy: a frequent cause of acute flaccid paralysis in China. *Ann Neurol* 1993;33:333–342
- Griffin JW, Li CY, Macko C, et al. Early nodal changes in the acute motor axonal neuropathy pattern of the Guillain-Barré syndrome. *J Neurocytol* 1996;25:33–51
- Ho TW, Mishu B, Li CY, et al. Guillain-Barré syndrome in northern China: relationship to *Campylobacter jejuni* infection and anti-glycolipid antibodies. *Brain* 1995;118:597–605
- Coe CJ. Guillain-Barré syndrome in Korean children. *Yonsei Med J* 1989;30:81–87
- Jackson CE, Barohn RJ, Mendell JR. Acute paralytic syndrome in three American men. Comparison with Chinese cases. *Arch Neurol* 1993;50:732–735
- Ramos-Alvarez M, Bessudo L, Sabin A. Paralytic syndromes associated with noninflammatory cytoplasmic or nuclear neuropathy: acute paralytic disease in Mexican children, neuropathologically distinguishable from Landry-Guillain-Barré syndrome. *JAMA* 1969;207:1481–1492
- Ho TW, Roberts M, Hsieh S-T, et al. Mechanism of paralysis and recovery in acute motor axonal neuropathy after *Campylobacter jejuni* infection. *Ann Neurol* 1995;38:350 (Abstract)
- Gupta SK, Taly AB, Suresh TG, et al. Acute idiopathic axonal neuropathy (AIAN): a clinical and electrophysiological observation. *Acta Neurol Scand* 1994;89:220–224
- Gregson NA, Jones D, Thomas PK, Willison HJ. Acute motor neuropathy with antibodies to GM1 ganglioside. *J Neurol* 1991;238:447–451
- Rees JH, Soudain SE, Gregson NA, Hughes RA. *Campylobacter jejuni* infection and Guillain-Barré syndrome. *N Engl J Med* 1995;333:1374–1379
- Yuki N, Yoshino H, Sato S, Miyatake T. Acute axonal polyneuropathy associated with anti-GM1 antibodies following *Campylobacter jejuni* enteritis. *Neurology* 1990;40:1900–1902
- Yuki N, Sato S, Fujimoto S, et al. Serotype of *Campylobacter*

Fig 3. (A and B) Light-electron microscopic (light-EM) comparisons of this fiber demonstrated C3d deposition on the nodal axolemma. A shows at the EM level the same node of Ranvier illustrated in Figure 2A; this node was intensely immunostained for C3d. At the EM level, much of the perimeter of this node is occupied only by the axolemma and the overlying Schwann cell basal lamina, without overlying Schwann cell cytoplasm. The macrophage (M) overlying this node of Ranvier is extending a process to the node. The boxed region in A is shown at higher power in B. Note that the macrophage process (M) has passed through the Schwann cell basal lamina and apposes the axon (A), displacing the overlying Schwann cell process. Images such as this demonstrate that the ring of immunostaining identified at the light level is associated with the axolemma, rather than Schwann cell cytoplasm over the node. (A, $\times 19,040$ before 7% reduction; B, $\times 109,500$ before 7% reduction.) (C) In this ventral root from an acute motor axonal neuropathy case at 5 days, a macrophage (M) has insinuated itself under the myelin terminal loops and is extending radially around the axon (A) within the periaxonal space. ($\times 9,000$ before 7% reduction.) (D) This fiber has a macrophage (M) within the periaxonal space of the myelinated internode. As shown in the inset, the fiber appears otherwise normal with no disruption of myelin. The axon (A) is compressed by the macrophage. This image corresponds to the type of periaxonal macrophages illustrated by light microscopic immunocytochemistry in Figure 2E and F. ($\times 13,040$ before 7% reduction; inset, $\times 1,620$ before 7% reduction.)

- jejuni*, HLA, and the Guillain-Barré syndrome. Muscle Nerve 1992;16:968–969
20. Fujimoto S, Yuki N, Itoh T, Amako K. Specific serotype of *Campylobacter jejuni* associated with Guillain-Barré syndrome. J Infect Dis 1992;165:183 (Letter)
 21. Kornberg AJ, Pestronk A, Bieser K, et al. The clinical correlates of high-titer IgG anti-GM1 antibodies. Ann Neurol 1994;35:234–237
 22. Rees JH, Gregson NA, Hughes RAC. Anti-ganglioside GM1 antibodies in Guillain-Barré syndrome and their relationship to *Campylobacter jejuni* infection. Ann Neurol 1995;38:809–816
 23. Ganser AL, Kirschner DA. Differential expression of gangliosides on the surfaces of myelinated nerve fibers. J Neurosci Res 1984;12:245–255
 24. Ganser AL, Kirschner DA, Willinger M. Ganglioside localization on myelinated nerve fibers by cholera toxin binding. J Neurocytol 1983;12:921–938
 25. Santoro M, Thomas FP, Fink ME, et al. IgM deposits at nodes of Ranvier in a patient with amyotrophic lateral sclerosis, anti-GM₁ antibodies, and multifocal motor conduction block. Ann Neurol 1990;28:373–377
 26. Thomas FP, Trojaborg W, Nagy C, et al. Experimental autoimmune neuropathy with anti-GM1 antibodies and immunoglobulin deposits at the nodes of Ranvier. Acta Neuropathol 1991;82:378–383.
 27. Yuki N, Handa S, Taki T, et al. Cross-reactive antigen between nervous tissue and a bacterium elicits Guillain-Barré syndrome: molecular mimicry between ganglioside GM1 and lipopolysaccharide from Penner's serotype 19 of *Campylobacter jejuni*. Bio-med Res 1992;13:451–453
 28. Yuki N, Taki T, Inagaki F, et al. A bacterium lipopolysaccharide that elicits Guillain-Barré syndrome has a GM1 ganglioside-like structure. J Exp Med 1993;178:1771–1775
 29. Yuki N. Pathogenesis of axonal Guillain-Barré syndrome: hypothesis. Muscle Nerve 1994;17:680–682
 30. Powell HC, Myers RR. The axon in Guillain-Barré syndrome: immune target or innocent bystander? Ann Neurol 1996;39:4–5 (Editorial)
 31. Stoll G, Griffin JW, Li CY, Trapp BD. Wallerian degeneration in the peripheral nervous system: participation of both Schwann cells and macrophages in myelin degradation. J Neurocytol 1989;18:671–683
 32. Hsu S-M, Raine L, Fanger H. Use of avidin-biotin-peroxidase complex (ABC) in immunoperoxidase techniques: a comparison between ABC and unlabeled antibody (PAP) procedures. J Histochem Cytochem 1981;29:577–580
 33. Law SK, Levine RP. Interaction between the third complement protein and cell surface macromolecules. Proc Natl Acad Sci USA 1977;74:2701–2705
 34. Lachmann PJ, Pangburn MK, Oldroyd RG. Breakdown of C3 after complement activation: identification of a new fragment, C3g, using monoclonal antibodies. J Exp Med 1982;156:205–216
 35. Trapp BD, Itoyama Y, Sternberger NH, et al. Immunocytochemical localization of P₀ protein in Golgi complex membranes and myelin of developing rat Schwann cells. J Cell Biol 1981;90:1–6
 36. Brechenmacher C, Vital C, Laurentjoly L, Castaing Y. Ultrastructural study of peripheral nerve in Guillain-Barré syndrome. Presence of mononuclear cells in axons. Acta Neuropathol 1981;7(suppl):249–251
 37. Gregson NA, Koblar S, Hughes RAC. Antibodies to gangliosides in Guillain-Barré syndrome: specificity and relationship to clinical features. Q J Med 1993;86:111–117
 38. Willison HJ, Paterson G, Kennedy PG, Veitch J. Cloning of human anti-GM1 antibodies from motor neuropathy patients. Ann Neurol 1994;35:471–478
 39. Yuki N, Taki T, Takahashi M, et al. Penner's serotype 4 of *Campylobacter jejuni* has a lipopolysaccharide that bears a GM1 ganglioside epitope as well as one that bears a GD1a epitope. Infect Immun 1994;62:2101–2103
 40. Thomas FP, Adapon HP, Goldberg GP, et al. Localization of neural epitopes that bind to IgM monoclonal autoantibodies (M-proteins) from two patients with motor neuron disease. J Neuroimmunol 1989;21:31–39
 41. Corbo M, Quattrini A, Latov N, Hays AP. Localization of GM1 and Gal(beta 1–3)GalNAc antigenic determinants in peripheral nerve. Neurology 1993;43:809–814
 42. Santoro M, Uncini A, Corbo M, et al. Experimental conduction block induced by serum from a patient with anti-GM1 antibodies. Ann Neurol 1992;31:385–390
 43. Illa I, Ortiz N, Gallard E, et al. Acute axonal Guillain-Barré syndrome with IgG antibodies against motor axons following parenteral gangliosides. Ann Neurol 1995;38:218–224
 44. Waxman SG. Sodium channel blockade by antibodies: a new mechanism of neurological disease? Ann Neurol 1995;37:436–442
 45. Takigawa T, Yasuda H, Kikkawa R. Antibodies against GM1 ganglioside affect K⁺ and Na⁺ currents in isolated rat myelinated nerve fibers. Ann Neurol 1995;37:436–442
 46. Roberts M, Willison H, Vincent A, Newsom-Davis J. Guillain-Barré syndrome and neurotransmitter release. Lancet 1994;343:454–455 (Abstract)
 47. Hirano A. Reaction of the periaxonal space to some pathological process. In: Zimmerman HM, ed. Progress in neuropathology, vol 5. New York: Raven Press, 1983:99–112
 48. Trapp BD, Quarles RH, Suzuki K. Immunocytochemical studies of quaking mice support a role for the myelin-associated glycoprotein in forming and maintaining the periaxonal space and periaxonal cytoplasmic collar of myelinating Schwann cells. J Cell Biol 1984;99:594–606